

You are allowed to use AI, consult friends, TAs, but just do yourself a favor and understand what you are writing. You are not allowed to post the assignment questions on any forums other than our Piazza.

Please do not ask instructors to help with assignment questions. This is to ensure pedagogical equity and encourage independent problem solving.

Every question on this assignment is worth 10 points.

- Q1.** Prove that a set is semi-decidable iff it is m -reducible to $H = \{(e, x) \in \mathbb{N}^2 : M_e(x) \text{ halts}\}$.
- Q2.** Prove that every infinite semi-decidable set contains an infinite decidable subset.
- Q3.** Prove that if a set A of natural numbers is semi-decidable and infinite, then there exists an injective computable function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $\text{Range}(f) = A$.
- Q4.** Many standard Gödel numbering schemes (such as those based on prime factorization or direct string encoding) are injective but not surjective – leaving “gaps” of natural numbers that do not correspond to valid Turing machine syntax. Prove that there exists a computable bijection between the set of Gödel numbers of (valid syntax) Turing machines and the natural numbers.
- Q5.** Prove that the set **Tot** = $\{e \in \mathbb{N} : M_e \text{ has domain } \mathbb{N}\}$ (i.e. the set of the indices of total computable functions) is not semi-decidable.
- Your solutions must assume towards a contradiction that **Tot** is semi-decidable. Then, show that this assumption implies the existence of some computable f such that $\{f(e) : e \in \mathbb{N}\}$ is the set of indices of computable functions. This will imply that the function given by $g(x) = M_{f(x)}(x) + 1$ is computable (explain why). Finally, find a contradiction.
- Q6.** Prove that the Predecessor function is primitive recursive.

Questions continue on the next page.

Q7. If $\mathcal{C} \subseteq \mathcal{P}(\mathbb{N})$, and $X \in \mathcal{C}$, we will say that X is $\mathcal{C}$ -complete if $\forall Y \in \mathcal{C}, Y \leq_m X$.

Prove that $\mathbf{Fin} = \{e \in \mathbb{N} : M_e \text{ halts on only finitely many elements}\}$ is Σ_2^0 -complete.

Q8. For A, B sets of natural numbers, we will write $A \leq_p B$ if $A \leq_m B$ via a reduction function that is computable in polynomial time. More clearly, $A \leq_p B$ means that:

There exists $f: \mathbb{N} \rightarrow \mathbb{N}$ such that f is computable in polynomial time, and

$$\forall x \in \mathbb{N}, x \in A \Leftrightarrow f(x) \in B.$$

This new reducibility is known as **polynomial-time** reduction, which is important in studying complexity classes.

Let $P = \{X \subseteq \mathbb{N} : X \text{ is decidable in polynomial time}\}$.

- (a) Let $U \in P$, and suppose that $X \leq_p U$. Prove that X must be in P .
- (b) Prove that, $\forall Y \in P, Y \leq_p \{1\}$.
- (c) [True or False] $\forall Y \in \Sigma_2^0, Y \leq_p \mathbf{Fin}$. Please justify.
- (d) [True or False] $\forall Y \in \Sigma_1^0, Y \leq_p H$ (the halting set). Please justify.

Q9. Bonus

The last question part (b) shows that $\leq_p$ trivializes the concept of completeness for the class P , which is a very important class in Complexity Theory. In practice, P -completeness is based on another type of reduction known as **Log-Space** reduction (denoted $\leq_L$). We write $A \leq_L B$ if the reduction function is computable in $O(\log n)$ space (excluding the space for the read-only input and write-only output). Read about Log-space reduction, and the Circuit Value Problem (CVP).

Prove that: CVP is P -complete (here P -completeness is with respect to $\leq_L$).

Q10. Bonus

Prove that, for any given $e \in \mathbf{Tot}$, there exists $n \in \mathbb{N}$ such that $M_{M_e(n)} = M_n$.